

FEC simulation — 25% overhead, EEP vs UEP (ViSQOL)

ViSQOL audio-mode score /100 (100 = MOS 4.732 ceiling), 30 s window · 0.5 redundant copies per chunk average, identical pool for both modes · EEP: content-blind uniform alternation · UEP: importance-ranked triage, silence skipped · Channels: Bernoulli (random) and Gilbert-Elliott bursts (mean 80 ms) · Listener = 0.6 × vocal + 0.4 × instrumental

lamour de ma vie

16,725 packets · 0.4% silent

Loss	Channel	Placement	Mode	V.rec	I.rec	V.qual	I.qual	Listener
1%	bernoulli	—	EEP	99.2%	99.2%	99.2	97.3	98.4
	bernoulli	—	UEP	99.4%	99.2%	98.4	96.6	97.7
	burst	continuous	EEP	99.2%	99.2%	99.7	99.6	99.7
	burst	continuous	UEP	99.0%	99.0%	97.6	97.7	97.6
	burst	staggered	EEP	99.3%	99.3%	99.6	99.6	99.6
	burst	staggered	UEP	99.4%	99.3%	99.3	99.0	99.2
5%	bernoulli	—	EEP	96.2%	96.3%	92.6	89.9	91.5
	bernoulli	—	UEP	96.8%	95.7%	94.9	91.5	93.5
	burst	continuous	EEP	94.7%	94.7%	94.3	94.2	94.3
	burst	continuous	UEP	94.9%	94.9%	94.9	95.8	95.3
	burst	staggered	EEP	96.4%	96.4%	96.6	96.2	96.4
	burst	staggered	UEP	96.5%	95.3%	96.2	94.2	95.4
10%	bernoulli	—	EEP	92.4%	92.2%	87.5	77.6	83.5
	bernoulli	—	UEP	93.3%	91.2%	81.8	78.5	80.5
	burst	continuous	EEP	90.1%	90.1%	92.2	84.4	89.1
	burst	continuous	UEP	90.8%	90.8%	89.4	84.6	87.5
	burst	staggered	EEP	90.9%	91.0%	93.4	89.7	91.9
	burst	staggered	UEP	93.3%	90.9%	92.8	88.8	91.2
20%	bernoulli	—	EEP	83.8%	83.6%	78.1	63.6	72.3
	bernoulli	—	UEP	86.4%	82.1%	71.4	58.5	66.2
	burst	continuous	EEP	80.5%	80.5%	83.7	80.1	82.3
	burst	continuous	UEP	79.4%	79.4%	83.1	76.4	80.4
	burst	staggered	EEP	83.3%	83.4%	82.3	73.2	78.7
	burst	staggered	UEP	85.2%	81.2%	82.8	76.9	80.4

man i need

10,075 packets · 9.6% silent — the most content asymmetry

Loss	Channel	Placement	Mode	V.rec	I.rec	V.qual	I.qual	Listener
1%	bernoulli	—	EEP	99.2%	99.2%	98.9	99.3	99.1
	bernoulli	—	UEP	99.4%	99.0%	98.8	99.2	99.0
	burst	continuous	EEP	98.9%	98.9%	99.2	99.4	99.3
	burst	continuous	UEP	99.3%	99.3%	99.0	99.3	99.1
	burst	staggered	EEP	99.2%	99.3%	99.2	99.4	99.3
	burst	staggered	UEP	99.7%	99.3%	99.9	100.0	99.9
5%	bernoulli	—	EEP	96.3%	95.9%	93.2	93.2	93.2
	bernoulli	—	UEP	97.1%	96.0%	96.4	94.5	95.6
	burst	continuous	EEP	94.7%	94.7%	97.6	97.7	97.6
	burst	continuous	UEP	96.0%	96.0%	96.7	97.0	96.8
	burst	staggered	EEP	95.7%	95.7%	96.9	97.7	97.2
	burst	staggered	UEP	97.0%	96.2%	98.2	96.3	97.4
10%	bernoulli	—	EEP	92.9%	92.5%	87.3	92.5	89.4
	bernoulli	—	UEP	94.7%	92.4%	93.4	90.0	92.0
	burst	continuous	EEP	89.4%	89.4%	89.9	90.0	89.9
	burst	continuous	UEP	90.7%	90.7%	87.8	87.1	87.5
	burst	staggered	EEP	92.8%	92.8%	91.7	91.8	91.7
	burst	staggered	UEP	94.0%	92.5%	93.9	91.6	93.0
20%	bernoulli	—	EEP	84.1%	83.7%	78.0	81.2	79.3
	bernoulli	—	UEP	88.5%	83.3%	86.7	80.4	84.2
	burst	continuous	EEP	79.7%	79.7%	76.5	77.8	77.0
	burst	continuous	UEP	83.1%	83.1%	78.9	79.0	78.9
	burst	staggered	EEP	83.3%	83.5%	80.4	81.6	80.9
	burst	staggered	UEP	87.0%	82.5%	87.8	84.1	86.3

the cure

14,750 packets · 0.1% silent

Loss	Channel	Placement	Mode	V.rec	I.rec	V.qual	I.qual	Listener
1%	bernoulli	—	EEP	99.2%	99.3%	98.9	98.6	98.8
	bernoulli	—	UEP	99.3%	99.2%	99.1	99.5	99.3
	burst	continuous	EEP	99.0%	99.0%	99.4	98.4	99.0
	burst	continuous	UEP	99.2%	99.2%	99.5	99.7	99.6
	burst	staggered	EEP	99.2%	99.2%	98.8	98.9	98.8
	burst	staggered	UEP	99.5%	99.3%	99.5	99.2	99.4
5%	bernoulli	—	EEP	96.3%	96.3%	94.5	93.4	94.1
	bernoulli	—	UEP	96.6%	96.2%	93.5	92.9	93.3
	burst	continuous	EEP	95.3%	95.3%	96.1	94.4	95.4
	burst	continuous	UEP	93.8%	93.8%	97.0	95.8	96.5
	burst	staggered	EEP	95.8%	95.7%	93.2	90.2	92.0
	burst	staggered	UEP	96.4%	95.8%	97.7	97.6	97.7
10%	bernoulli	—	EEP	92.6%	92.2%	89.4	86.2	88.1
	bernoulli	—	UEP	92.6%	91.9%	87.9	88.3	88.1
	burst	continuous	EEP	89.7%	89.7%	91.5	86.4	89.5
	burst	continuous	UEP	90.2%	90.2%	91.7	88.5	90.4
	burst	staggered	EEP	90.8%	90.7%	89.7	87.3	88.7
	burst	staggered	UEP	92.4%	91.5%	92.0	91.3	91.7
20%	bernoulli	—	EEP	84.3%	83.7%	80.6	71.8	77.1
	bernoulli	—	UEP	85.0%	83.2%	78.6	77.2	78.0
	burst	continuous	EEP	79.3%	79.3%	79.6	69.7	75.6
	burst	continuous	UEP	81.1%	81.1%	75.3	70.5	73.4
	burst	staggered	EEP	81.9%	81.8%	84.0	75.7	80.7
	burst	staggered	UEP	84.4%	82.7%	88.3	86.2	87.5

green = UEP beats EEP by at least 0.5 listener points in the same condition · red = UEP loses to EEP by at least 0.5